

Q:1 Modify Bellman-Ford so that after each pass over all edges, we check whether any edges were relaxed. If in some pass no relaxation occurs, terminate immediately. After k passes, all shortest paths of at most k edges have been computed. Since the shortest path to any vertex uses at most $(n-1)$ edges, all distances stabilize by the $(n+1)$ -th pass. At this point, no relaxation occurs, so the algorithm halts automatically without knowing in advance.

During each pass:

changed = false

for each edge (u, v) in E :

if $\text{list}[u] + w(u, v) < \text{list}[v]$:

$\text{list}[v] = \text{list}[u] + w(u, v)$

changed = true

if changed == false:

terminate early // no need to run remaining passes.

Q:2 Topological order: x, s, t, x, y, z

Dry Run:-

$\text{list}(x) = 0, \text{list}(s) = \infty, \text{list}(t) = \infty, \text{list}(x) = \infty,$

$\text{list}(y) = \infty, \text{list}(z) = \infty$

$\pi[x]$ to keep track of predecessor paths.

1) Process x :

• relax $x \rightarrow s$: $\text{dist}(s) = 0 + 5 = 5, \pi(s) = x$

• relax $x \rightarrow t$: $\text{dist}(t) = 0 + 3 = 3, \pi(t) = x$

After x :

$\text{dist} = \{x: 0, s: 5, t: 3, x: \infty, y: \infty, z: \infty\}$

2. Process s :

• relax $s \rightarrow t$: $5 + 2 = 7$ (no change, $\text{dist}(t) = 3$ is min).

• relax $s \rightarrow u$: $5 + 6 = 11 \rightarrow \text{dist}(u) = 11, \pi(u) = s$

After s:

$$\text{dist} = \{x=0, s=5, t=3, x=11, y=\infty, z=\infty\}$$

3) Process t:

- relax $t \rightarrow u = 3+7 = 10 \rightarrow$ improves x : $\text{dist}(x) = 10$, $\pi(x) = t$
- relax $t \rightarrow y = 3+4 = 7 \rightarrow \text{dist}(y) = 7$, $\pi(y) = t$
- relax $t \rightarrow z = 3+2 = 5 \rightarrow \text{dist}(z) = 5$, $\pi(z) = t$

After t:

$$\text{dist} = \{x=0, s=5, t=3, x=10, y=7, z=5\}$$

4) Process x:

- relax $x \rightarrow y = 10+(-1) = 9$ (no change, $\text{dist}(y) = 7$ is min)
- relax $x \rightarrow z = 10+1 = 11$ (no change, $\text{dist}(z) = 5$ is min)

After x:

$$\text{dist} = \{x=0, s=5, t=3, x=10, y=7, z=5\}$$

5) Process y:

- relax $y \rightarrow z = 7+(-2) = 5$ (no change, $\text{dist}(z) = 5$ is same)

After y: $\text{dist} = \text{unchanged}$

6) Process z:

- no outgoing edges. Therefore done!

Final shortest-path distances from s:

- $\text{dist}(s) = 0$
- $\text{dist}(s) = 5$ (path: $s \rightarrow s$)
- $\text{dist}(t) = 3$ (path: $s \rightarrow t$)
- $\text{dist}(x) = 10$ (path: $s \rightarrow t \rightarrow u$)
- $\text{dist}(y) = 7$ (path: $s \rightarrow t \rightarrow y$)
- $\text{dist}(z) = 5$ (path: $s \rightarrow t \rightarrow y \rightarrow z$) or (path: $s \rightarrow t \rightarrow z$)

$\therefore s \rightarrow t \rightarrow z (5)$, $s \rightarrow t \rightarrow y \rightarrow z (5)$.

$\Rightarrow$ Complexity Analysis:-

- 1. Topologic sort: $O(V+E)$

2. The relaxation loop, for each vertex (in topo order) relax every edge (u, v) once.
 - total work is $O(E)$.

so total running time = $O(V+E)$ space is $O(V)$
 for dist & π plus adjacency representation.

Q: 3 Source = S

initialization:

$$d(s) = 0, d(t) = d(x) = d(y) = d(z) = \infty$$

all $\pi = \text{NULL}$

We extract min & relax each outgoing edges in each iteration =

1) extract $u = s$

$$\cdot \text{relax } s \rightarrow t : d(t) = 10, \pi(t) = s$$

$$\cdot \text{relax } s \rightarrow y : d(y) = 5, \pi(y) = s$$

$$d = \{s=0, t=10, x=\infty, y=5, z=\infty\}$$

$$\pi = \{s:-, t:s, x:-, y:s, z:-\}$$

$$S = \{s\}$$

2) Extract $v = y$ ($d = 5$)

$$\cdot \text{relax } y \rightarrow t : d(t) = 5+3 = 8 \rightarrow \text{improves } t \text{ } [\pi(t) = y]$$

$$\cdot \text{relax } y \rightarrow x : d(x) = 5+9 = 14 \Rightarrow u = 14, \pi(x) = y$$

$$\cdot \text{relax } y \rightarrow z : d(z) = 5+2 = 7 = z = 7, \pi(z) = y$$

$$d = \{s=0, t=8, x=14, y=5, z=7\}$$

$$\pi = \{s:-, t=y, x=y, y:s, z=y\}$$

$$S = \{s, y\}$$

3) Extract $u = z$ ($d = 7$)

$$\cdot \text{relax } z \rightarrow x : d(x) = 7+6 = 13 \rightarrow \text{improves } x \text{ } [\pi(x) = z]$$

$$\cdot \text{relax } z \rightarrow s : d(s) = 7+7 = 14 \rightarrow \text{no improvement on } s.$$

$$d = \{s=0, t=8, x=13, y=5, z=7\}$$

$$\pi = \{s:-, t=y, x=z, y=s, z=y\}$$

$$S = \{s, y, z\}$$

4) Extract $u=t$ ($d=8$)

- relax $t \rightarrow x : d(x) = 8+1 = 9 \rightarrow$ improves x [$\pi(x)=t$]
- relax $t \rightarrow y : d(y) = 8+2 = 10 \rightarrow$ no improvement for y as $10 > 5$

$$d = \{s=0, t=8, x=9, y=5, z=7\}$$

$$\pi = \{s:-, t:y, x=t, y=5, z=y\}$$

$$s = \{s, y, z, t\}$$

5) Extract $u=x$ ($d=9$)

- relax $x \rightarrow z : d(z) = 9+4 = 13 \rightarrow$ no improvement in z as $13 > 7$

$$d = \{s=0, t=8, x=9, y=5, z=7\}$$

$$\pi = \{s:-, x=t, y=5, z=y\}$$

$$s = \{s, y, z, t, x\}$$

shortest distance froms:-

$$d(s) = 0$$

$$d(y) = 5 \text{ via } s \rightarrow y$$

$$d(z) = 7 \text{ via } s \rightarrow y \rightarrow z$$

$$d(t) = 8 \text{ via } s \rightarrow y \rightarrow t$$

$$d(x) = 9 \text{ via } s \rightarrow y \rightarrow t \rightarrow x$$

source = s

initialize :

$$d(z) = 0, d(x) = d(y) = d(t) = d(s) = \infty, \pi = \text{Null}$$

1) Extract $u=z$ ($d=0$)

- $z \rightarrow x = 0+6 = 6 \rightarrow d(x) = 6$, improves x [$\pi(x)=z$]
- $z \rightarrow s = 0+7 = 7 \rightarrow d(s) = 7$, improves s [$\pi(s)=z$]

$$d = \{z=0, x=6, s=7, y=\infty, t=\infty\}$$

$$\pi = \{z:-, x=z, s=z, y=-, t=-\}$$

$$s = \{z\}$$

2) Extract $u=x$ ($d=6$)

- $x \rightarrow z = 6+4 = 10 \rightarrow$ no improvement for z as $10 > 0$

$$d = \{z: 0, x: 6, y: \infty, t: \infty\}$$

$$\pi = \{z: -, x: z, s: z, y: -, t: -\}$$

$$S = \{z, x\}$$

3) Extract $u = s$ ($d = 7$)

$$s \rightarrow t: 7 + 10 = 17 \rightarrow d(t) = 17, \text{ improves } t \quad [\pi(t) = s]$$

$$s \rightarrow y: 7 + 5 = 12 \rightarrow d(y) = 12, \text{ improves } y \quad [\pi(y) = s]$$

$$d = \{z: 0, x: 6, s: 7, y: 12, t: 17\}$$

$$\pi = \{z: -, x: z, s: z, y: s, t: s\}$$

$$S = \{z, x, s\}$$

4) Extract $u = y$ ($d = 12$)

$$y \rightarrow t = 12 + 3 = 15 \rightarrow d(t) = 15, \text{ improves } t \quad [\pi(t) = y]$$

$$y \rightarrow x = 12 + 9 = 21 \rightarrow \text{no improvement for } x \text{ as } 21 > 6$$

$$y \rightarrow z = 12 + 2 = 14 \rightarrow \text{no improvement for } z \text{ as } 14 > 0$$

$$d = \{z: 0, x: 6, s: 7, y: 12, t: 15\}$$

$$\pi = \{z: -, x: z, s: z, y: s, t: y\}$$

$$S = \{z, x, s, y\}$$

5) Extract $u = t$ ($d = 15$)

$$\text{relax } t \rightarrow y = 15 + 2 = 17 \rightarrow \text{no improvement for } y \text{ since } 17 > 12$$

$$\text{relax } t \rightarrow x = 15 + 1 = 16 \rightarrow \text{no improvement for } x \text{ since } 16 > 6$$

$$d = \{z: 0, x: 6, s: 7, y: 12, t: 15\}$$

$$\pi = \{z: -, x: z, s: z, y: s, t: y\}$$

$$S = \{z, x, s, y, t\}$$

find shortest distance from z :

$$d(z) = 0$$

$$d(x) = 6 \quad \text{via } z \rightarrow x$$

$$d(s) = 7 \quad \text{via } z \rightarrow s$$

$$d(y) = 12 \quad \text{via } z \rightarrow s \rightarrow y$$

$$d(t) = 15 \quad \text{via } z \rightarrow s \rightarrow y \rightarrow t$$

Q.4

	1	2	3	4	5	6	
$D^0 =$	1	0	∞	∞	∞	-1	∞
	2	1	0	∞	2	∞	∞
	3	∞	2	0	∞	∞	$-\infty$
	4	-4	∞	∞	0	3	∞
	5	∞	7	∞	∞	0	∞
	6	∞	5	10	∞	∞	0

	1	2	3	4	5	6	
$D^1 =$	1	0	∞	∞	∞	-1	∞
2	2	2	2	2	0	∞	
3	∞	2	0	∞	∞	$-\infty$	
4	-4	∞	∞	0	-5	∞	
5	∞	7	∞	∞	0	∞	
6	∞	5	10	∞	∞	0	

$$\min \left(\begin{matrix} \infty, & 1 & + & \infty \\ D^0[2,3], & D^0[2,1] & + & D^0[1,3] \end{matrix} \right)$$

$$\min \left(\begin{matrix} 2 & 1 & + & \infty \\ D^0[2,x], & D^0[2,1] & + & D^0[1,x] \end{matrix} \right)$$

$$\min \left(\begin{matrix} \infty & 1 & + & \infty \\ D^0[2,5], & D^0[2,1] & + & D^0[1,5] \end{matrix} \right)$$

$$\infty, \quad 1 + (-1)$$

$$\min \left(\begin{matrix} D^0[3,x], & D^0[3,1] & + & D^0[1,x] \end{matrix} \right)$$

$$\min \left(\begin{matrix} D^0[3,5], & D^0[3,1] & + & D^0[1,5] \end{matrix} \right)$$

$$\infty, \quad \infty$$

$$\min \left(\begin{matrix} D^0[3,6], & D^0[3,1] & + & D^0[1,6] \end{matrix} \right)$$

$$\min \left(\begin{matrix} D^0[x,2], & D^0[x,3] & + & D^0[1,2] \end{matrix} \right)$$

$$\infty, \quad -x + \infty$$

	1	2	3	4	5	6
$D^2 = 1$	0	∞	∞	∞	-1	∞
2	1	0	∞	2	0	∞
3	3	2	0	4	2	-8
4	-4	∞	∞	0	-5	∞
5	8	7	∞	9	0	∞
6	6	5	10	7	5	0

	1	2	3	4	5	6		1	2	3	4	5	6
$D^3 = 1$	0	∞	∞	∞	-1	∞	$D^4 = 1$	0	∞	∞	∞	-1	∞
2	-2	0	∞	2	-3	∞	2	-2	0	∞	2	-3	∞
3	0	2	0	4	-1	-8	3	0	2	0	4	-1	-8
4	-4	∞	∞	0	-5	∞	4	-4	∞	∞	0	-5	∞
5	5	7	∞	9	0	∞	5	5	7	∞	9	0	∞
6	3	5	10	7	2	0	6	3	5	10	7	2	0

	1	2	3	4	5	6		1	2	3	4	5	6
$D^5 = 1$	0	6	∞	8	-1	∞	$D^6 = 1$	0	6	∞	8	-1	∞
2	-2	0	∞	2	3	∞	2	-2	0	∞	2	-3	∞
3	0	2	0	4	-1	∞	3	-5	-3	0	-1	-6	-8
4	4	2	∞	0	5	∞	4	-4	2	∞	0	-5	∞
5	5	7	∞	9	0	∞	5	5	7	∞	9	0	∞
6	3	5	10	7	2	0	6	3	5	10	7	2	0

Shortest path:

$1 \rightarrow 1 = 0$	$2 \rightarrow 1 = -2$	$3 \rightarrow 1 = -5$	$4 \rightarrow 1 = -4$
$1 \rightarrow 2 = 6$	$2 \rightarrow 2 = 0$	$3 \rightarrow 2 = -3$	$4 \rightarrow 2 = 2$
$1 \rightarrow 3 = \text{no path}$	$2 \rightarrow 3 = \text{no path}$	$3 \rightarrow 3 = 0$	$4 \rightarrow 3 = \text{no path}$
$1 \rightarrow 4 = 8$	$2 \rightarrow 4 = 2$	$3 \rightarrow 4 = -1$	$4 \rightarrow 4 = 0$
$1 \rightarrow 5 = -1$	$2 \rightarrow 5 = -3$	$3 \rightarrow 5 = -6$	$4 \rightarrow 5 = -5$
$1 \rightarrow 6 = \text{no path}$	$2 \rightarrow 6 = \text{no path}$	$3 \rightarrow 6 = -8$	$4 \rightarrow 6 = \text{no path}$

$$5 \rightarrow 1 = 5 \quad 6 \rightarrow 1 = 3$$

$$5 \rightarrow 2 = 7 \quad 6 \rightarrow 2 = 5$$

$$5 \rightarrow 3 = \text{no path} \quad 6 \rightarrow 3 = 10$$

$$5 \rightarrow 4 = 8 \quad 6 \rightarrow 4 = 7$$

$$5 \rightarrow 5 = 0 \quad 6 \rightarrow 5 = 2$$

$$5 \rightarrow 6 = \text{no path} \quad 6 \rightarrow 6 = 0$$

	1	2	3	4	5
$D^0 = 1$	0	3	8	∞	-4
2	∞	0	∞	1	7
3	∞	4	0	∞	∞
4	2	∞	-5	0	∞
5	∞	∞	∞	6	0

	1	2	3	4	5
$D^1 = 1$	0	3	8	4	-4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

	1	2	3	4	5
$D^2 = 1$	0	3	-1	4	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

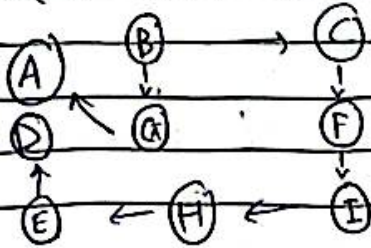
	1	2	3	4	5
$D^3 = 1$	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

	1	2	3	4	5
$D^4 = 1$	0	1	-3	2	-4
2	3	0	-4	1	-1
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

Shortest path:

$1 \rightarrow 1 = 0$	$2 \rightarrow 1 = 3$	$3 \rightarrow 1 = 7$	$4 \rightarrow 1 = 2$	$5 \rightarrow 1 = 8$
$1 \rightarrow 2 = 1$	$2 \rightarrow 2 = 0$	$3 \rightarrow 2 = 4$	$4 \rightarrow 2 = -1$	$5 \rightarrow 2 = 5$
$1 \rightarrow 3 = -3$	$2 \rightarrow 3 = -4$	$3 \rightarrow 3 = 0$	$4 \rightarrow 3 = -5$	$5 \rightarrow 3 = 1$
$1 \rightarrow 4 = 2$	$2 \rightarrow 4 = 1$	$3 \rightarrow 4 = 5$	$4 \rightarrow 4 = 0$	$5 \rightarrow 4 = 6$
$1 \rightarrow 5 = -4$	$2 \rightarrow 5 = 1$	$3 \rightarrow 5 = 3$	$4 \rightarrow 5 = -2$	$5 \rightarrow 5 = 0$

Q: 5(a) DFS: B, C, F, I, H, G, D, E, A



Visited Order = B, C, F, I, H, G, D, E, A

Steps:-

1. Start from B (t=1) $\rightarrow$ stack = [B] $\rightarrow$ visited = [B]
2. Choose C (t=2) $\rightarrow$ stack = [B, C] $\rightarrow$ visited = [B, C]
3. goes to F (t=3) $\rightarrow$ stack = [B, C, F] $\rightarrow$ visited = [B, C, F]
4. goes to I (t=4) $\rightarrow$ stack = [B, C, F, I] $\rightarrow$ visited = [B, C, F, I]
5. goes to H (t=5) $\rightarrow$ stack = [B, C, F, I, H] $\rightarrow$ visited = [B, C, F, I, H]
6. Choose G (t=6) $\rightarrow$ stack = [B, C, F, I, H, G] $\rightarrow$ visited = [B, C, F, I, H, G]
7. goes to D (t=7) $\rightarrow$ stack = [B, C, F, I, H, G, D] $\rightarrow$ visited = [B, C, F, I, H, G, D]
8. Pop D (t=8) $\rightarrow$ stack = [B, C, F, I, H, G] (D already in stack $\rightarrow$ no other option)
9. Pop G (t=9) $\rightarrow$ stack = [B, C, F, I, H]
10. Pop H (t=10) $\rightarrow$ stack = [B, C, F, I], visited = [B, C, F, I, H, G, D]
11. Pop I (t=11) $\rightarrow$ stack = [B, C, F]
12. Pop F (t=12) $\rightarrow$ stack = [B, C]
13. Pop C (t=13) $\rightarrow$ stack = [B]
14. goes to E (t=14) $\rightarrow$ stack = [B, E] $\rightarrow$ visited = [B, C, F, I, H, G, D, E]
15. Choose A (t=15) $\rightarrow$ stack = [B, E, A] [E had 2 direction only]
16. Pop A (t=16) $\rightarrow$ stack = [B, E] $\rightarrow$ visited = [B, C, F, I, H, G, D, E, A]
17. Pop E (t=17) $\rightarrow$ stack = [B]
18. Pop B (t=18) $\rightarrow$ stack = []

So DFS = B, C, F, I, H, G, D, E, A

Vertex = shortest time / finish time.

A : 15 / 16

F : 3 / 12

B : 1 / 18

G : 6 / 9

C : 2 / 13

H : 5 / 10

D : 7 / 8

I : 4 / 11

E : 14 / 17

b) Back Edges	Front Edges	Cross edges
$A \rightarrow B$	none b/c all	$A \rightarrow D$
$D \rightarrow H$	disjoint edges	$E \rightarrow H$
$H \rightarrow F$	tree edges	
$H \rightarrow T$		

c) SCC: {B, A, E}

Cycle: $B \rightarrow E \rightarrow A \rightarrow B$

SCC: {C}

(C has no path back from lower SCC)

SCC₃ = {F, I, H, G, D}

C₁ 4 mutually reachable cycles: $H \rightarrow G \rightarrow D \rightarrow H$, $H \rightarrow I$, $H \rightarrow F \rightarrow T \rightarrow H$, etc.

• Component Graph =

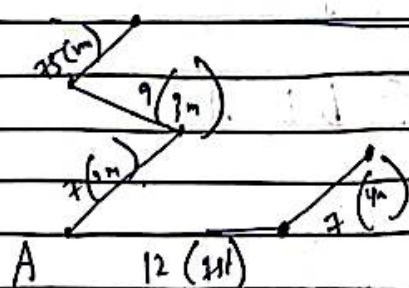
SCC₁ $\rightarrow$ SCC₂ (b/c $B \rightarrow C$)

SCC₁ $\rightarrow$ SCC₃ (b/c $E \rightarrow H$ & $A \rightarrow D$)

SCC₂ $\rightarrow$ SCC₃ (b/c $C \rightarrow F$)

Diagram = SCC₁ (B, A, E) $\rightarrow$ SCC₂ $\rightarrow$ SCC₃ (F, I, H, G, D)

Q: G(a)

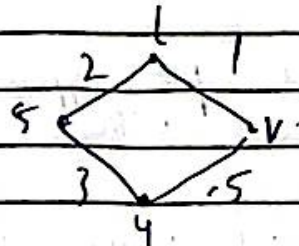


maximum spanning tree = Yes.

b) Yes, Prim's & Kruskal's algos can yield different maximum spanning trees if the graph has edges

with equal weights. However, the total weight of the maximum spanning tree will always be the same, if all edge weights are distinct, both algos produce the same tree.

Q: 7(a)



Connect shortest path from s to t = $s \rightarrow u \rightarrow v \rightarrow t$
 total weight = $3 + (-5) + 1 = -1$

$d(s) = 0$, $d(t) = 2$, $d(u) = 3$, $d(v) = \infty$

step	N'	u	t	v
0	s	3, 5	2, 5	∞
1	st	3, 5	"	∞
2	stu			-2, v
3	stuv			

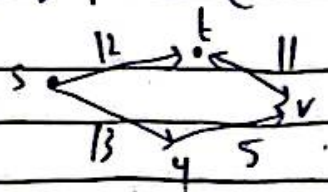
smallest tentative distance is $d(t) = 2$

later when u & v are on the path cost becomes -1, let has already finalized it, Hence, bijorker notes 2 instead of -1, has falls in presence of weight edges.

b) Counter example (using $k=10$)

Edge	Original	After adding k
$s \rightarrow t$	2	12
$s \rightarrow u$	3	13
$u \rightarrow v$	-5	5
$v \rightarrow t$	1	11

now the graph :- (After adding k)



Direct from s to $t = 12 (s \rightarrow t)$

Longer path $= s \rightarrow u \rightarrow v \rightarrow t = 13 + 4 + 11 = 28$

$\therefore$ after adding k , the direct path appears shorter, while originally the multi-edge way was better.

To conclude, this method is not valid b/c adding a constant changes relative path weights and not edges $\therefore$ can change the shortest path.

c) All shortest paths start with a single step from s only the first edge can be negative. After that, all edges are positive $\rightarrow$ Dijkstra works correctly b/c after the first hop from s , the remaining graph behaves like a standard positive-weight graph. Dijkstra finalizes nodes in order of correct increasing distances. therefore Dijkstra will not fail.

Proof :-

let $s \rightarrow x \rightarrow \dots \rightarrow u$ be the shortest path to u . After reaching x (in correct cost), the remaining path has only positive edges. Dijkstra will correctly relax paths from x to u , hence no incorrect finalization occurs.